ELEC 342

Lab Experiment: 2

Section: L1B

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Task 6A:

- We used this transformer as a step down transformer
- Using the results found in the open circuit test we can determine the value of parameters Rc,Xm

Open Circuit Test (Secondary)	Short Circuit Test (Primary)
Voc = 27.72V	Vsc = 3.25V
Ioc = 0.49A	Isc = 2.09A
Poc = 1.7W	Psc = 6W

$$PF = cos(\Theta) = \frac{Poc}{VocIoc}$$

$$Ye = \frac{Ioc}{Voc} <- cos^{-1}(PF)$$

$$Ye = \frac{1}{Rc} - j\frac{1}{Xm}$$

Rc = 521.3Ω
Xm = 61.1Ω

- Next, using the results found in the short circuit test we can determine the value of parameters Rsc,Xsc

$$PF = cos(\Theta) = \frac{Psc}{VscIsc}$$

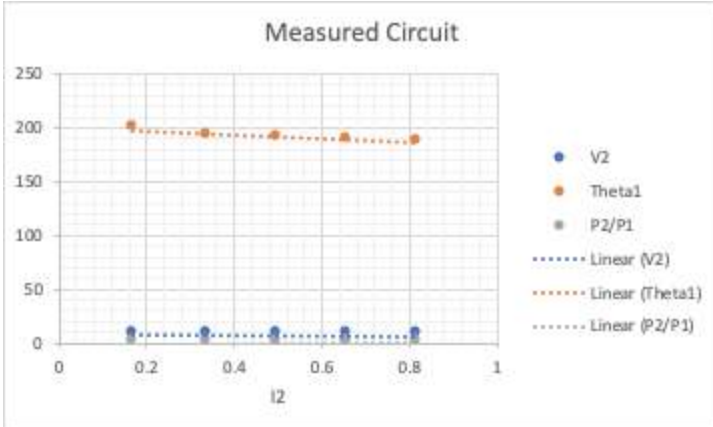
$$Ze = \frac{Isc}{Vsc} <- cos^{-1}(PF)$$

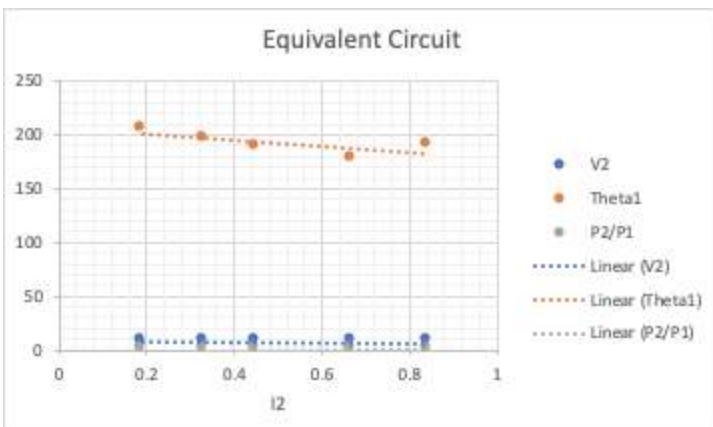
$$Ze = Rsc - jXsc$$

Rsc = 0.321Ω
Xsc = 0.158Ω

Task 6B:

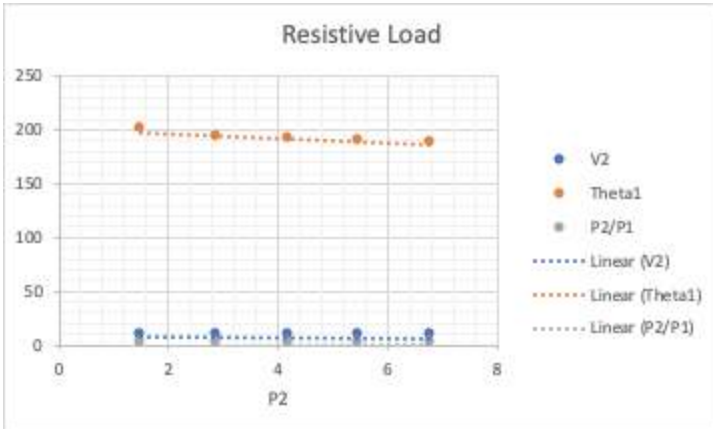
- It can be seen from these plots that the equivalent circuit is a very good approximation of the actual physical circuit. It predicts the loading characteristics very well.

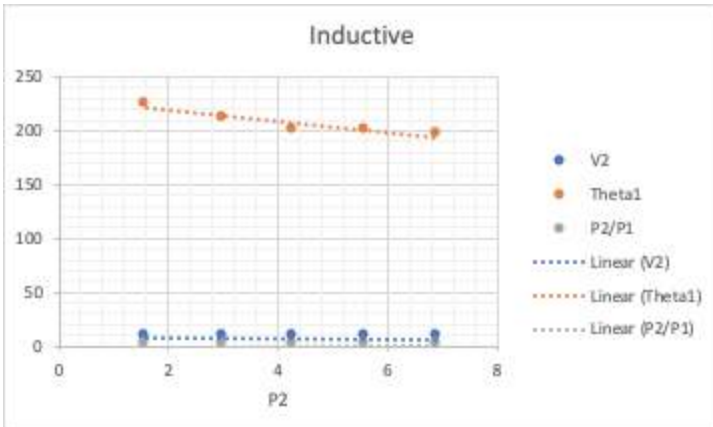


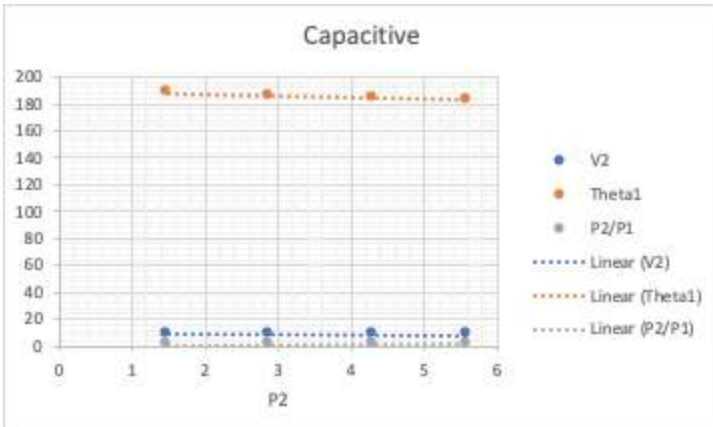


Task 6C:

- From the data we collected, the output voltage and real power did not change much when we added inductive and capacitive load. The real power results contradict what should happen in theory. The real power and voltage should decrease when we add inductive or capacitive load. To boost the real power and voltage, we should add inductors to a capacitive load or add capacitors to an inductive load.





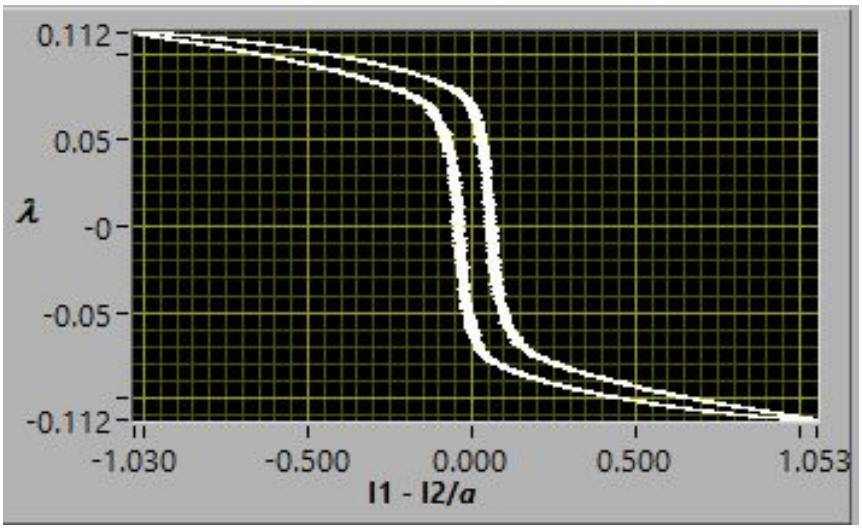


Task 6D:

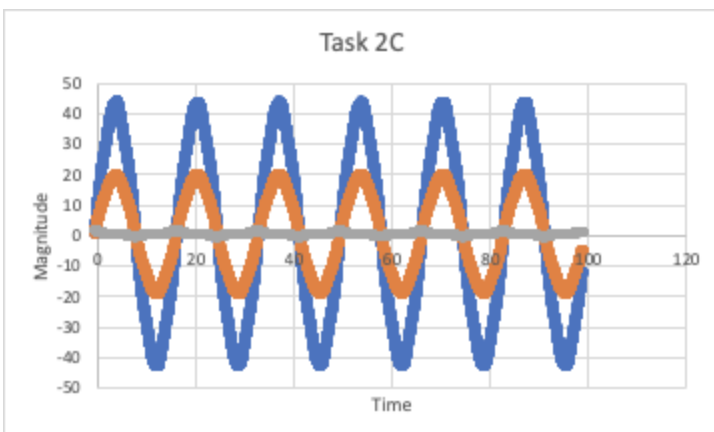
- Using the following equations we are able to obtain a value for the flux linkage

$$V1 = i1R1 + \frac{d\lambda}{dt}$$

$$\lambda(t) = \int (V1 - i1R1)dt + C\lambda$$



Hysteresis Loop



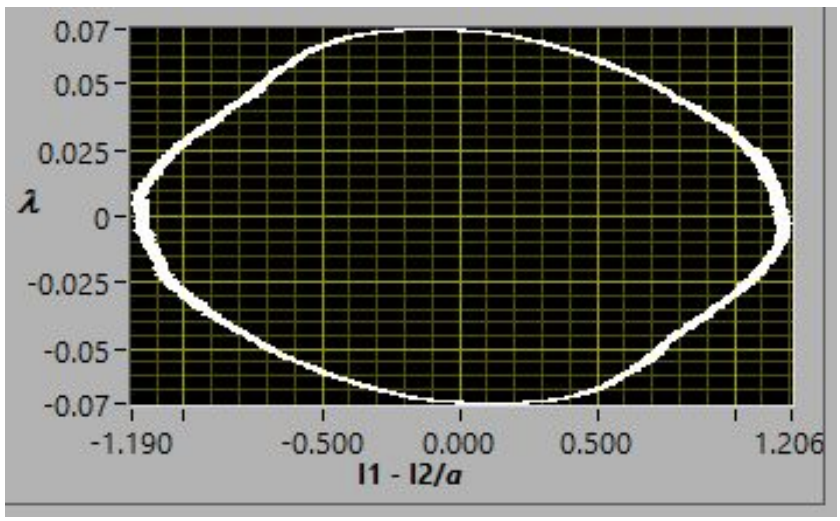
FFT Analysis

- Because of the nature of hysteresis, there is an oscillating nature to the core of a transformer. The rapid magnetization and demagnetization of the core due to changing currents results in loss for the core and also produced harmonics. The harmonically induced losses as a result are obviously not ideal in the physical

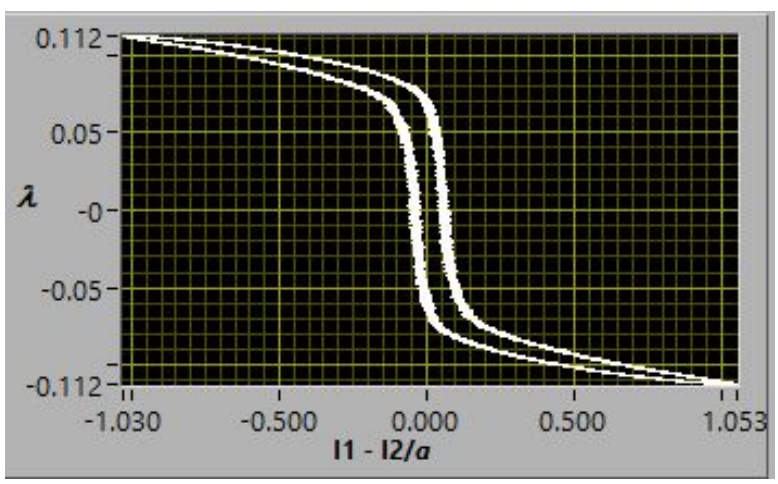
machine and represent one of the two central reasons for power loss in a transformer. The other being “eddy” currents due to Faraday’s Law.

Task 6E:

- There are two main sources of loss in the transformer core. The first is due to parasitic “eddy” currents that are a direct result of faraday’s law. They can heat up the container surrounding the transformer and cause serious damage to materials not built to withstand high currents. The other cause of major loss is due to hysteresis. Hysteresis losses are the result of the oscillating magnetization and demagnetization of the core as current flows. The hysteresis losses are directly proportional to the area of the λ-i1 loop. These two waveforms can help illustrate the difference.



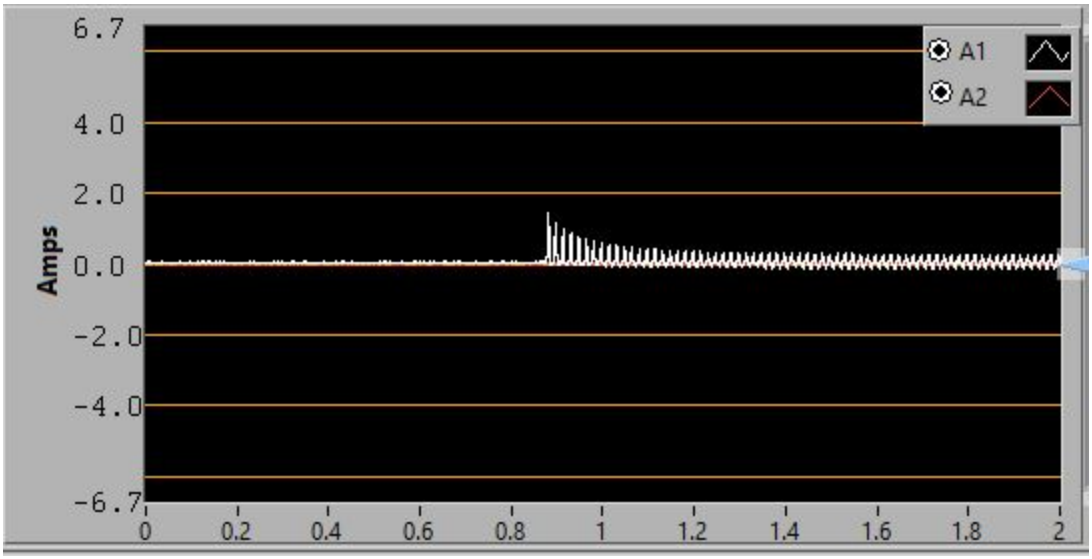
Large Hysteresis Losses



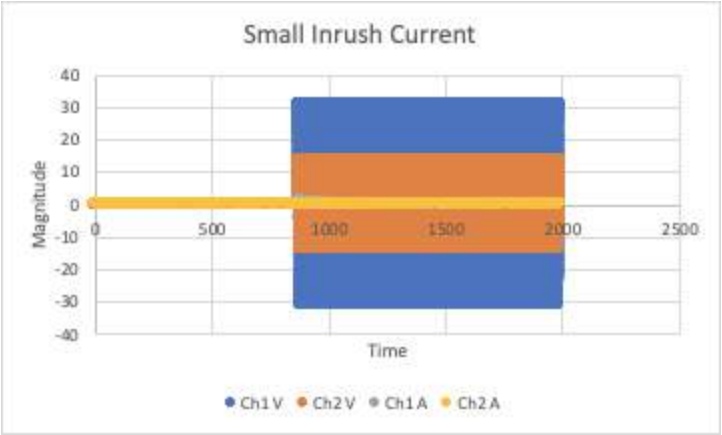
Small Hysteresis Losses

Task 6F:

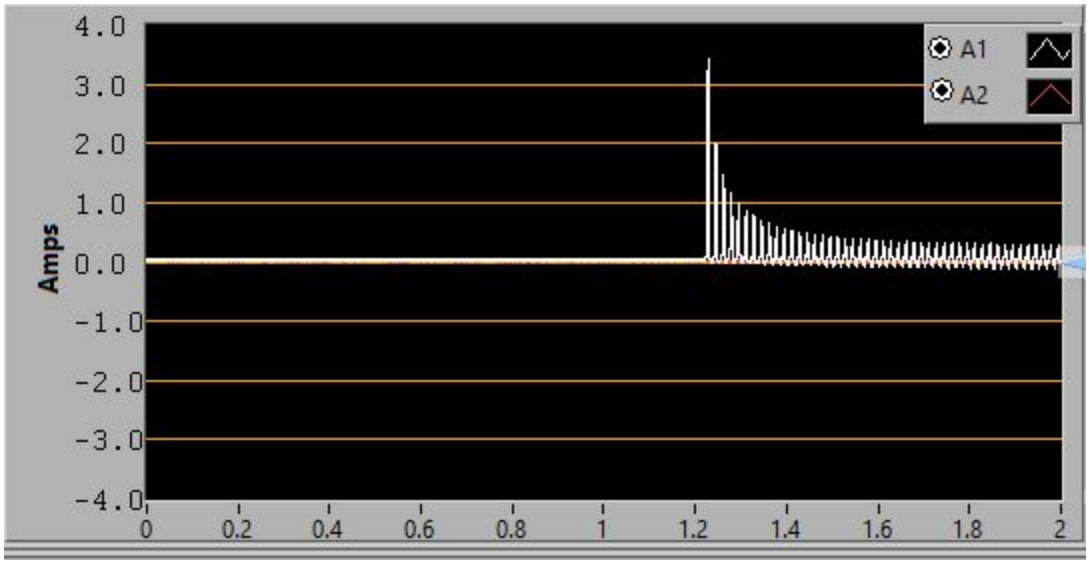
- By analyzing the waveforms and plots for the various voltages and currents for the case corresponding to the very high peak of the inrush current, it is clear that the case with the small inrush current is more favourable. The inrush current is the current that occurs during the energization of the transformer when the transformer is powering up. This current is usually much larger than the rated current and so it saturates the core. This is problematic for large transformers and can have crippling effects on the power network.



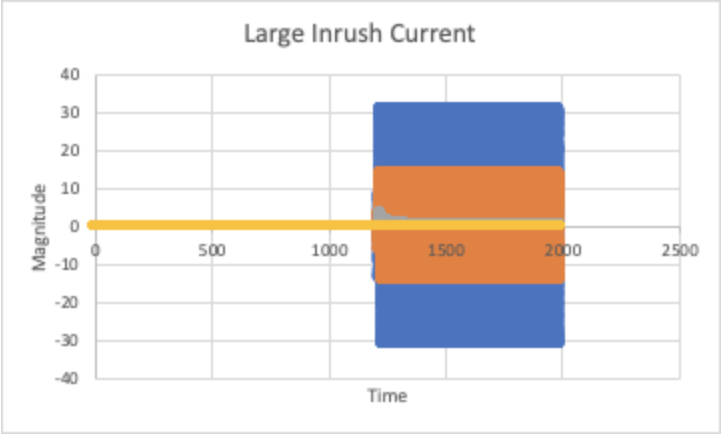
Small Inrush Current Waveform



Small Inrush Current Plot



Large Inrush Current Waveform



Large Inrush Current Plot

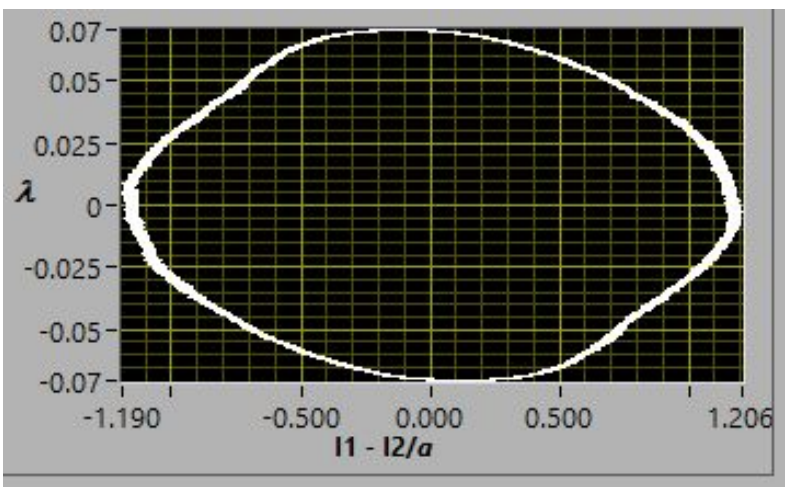
Task 6G:

(a) **Why do you think it is not a good idea to operate a transformer above its nominal (rated) voltage?**

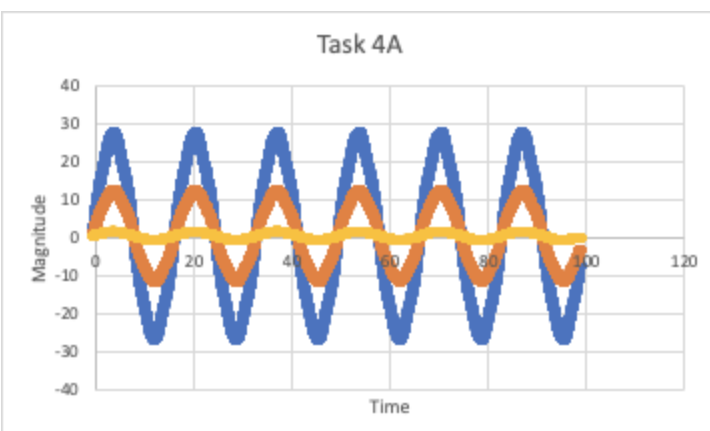
- It is not a good idea to operate the transformer above its nominated (rated) value because you risk saturation of the core. When the core is saturated, there will be a non-linear relationship between the applied voltage and the output stage which is clearly not ideal for measurements.

(b) **How can you describe the waveform of the primary current i1(t) when the transformer is loaded as in Task 4A (last measurement)? How this waveform is different from i1(t) under the no load case in Task 2A and C?**

- It can be seen from the hysteresis loop waveform that the primary current has a very wide area that the loop encompasses. This is related to the hysteresis losses, when compared to the unloaded case in part 2A and C the waveform is much narrower.



Hysteresis Loop



(c) **What can you say about the accuracy of the equivalent circuit? Does it predict all of the phenomena that you observed during this lab experiment?**

- After comparing results from measured values in the laboratory with the equivalent circuit composed of calculated circuit parameters, we come to the conclusion that this model is a good approximation of reality. The examined and calculated results were in support of one another and this result gives us further confidence in the utility of the equivalent circuit model.

(d) **Assuming you have a two-winding transformer that you have used in this lab, sketch the step up and step down connections. In each case, label the input and output voltages based on the nominal voltages found in Task 2.**

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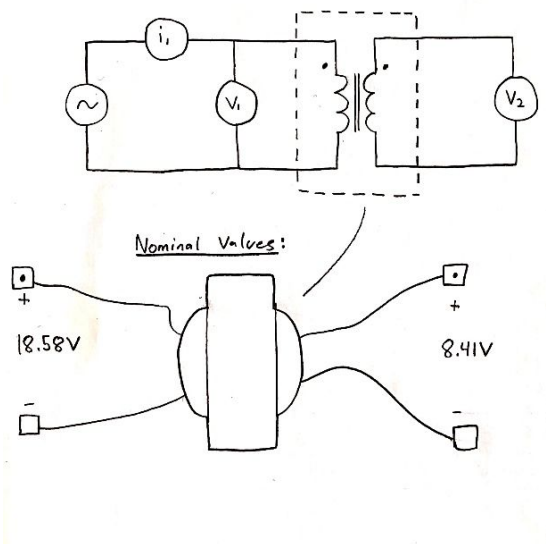

Summary:

- The objectives of this lab were to experiment and analyze a small two winding transformer by identifying the various circuit parameters and representing them in our equivalent circuit model. We also looked at the basic properties of practical single phase transformers and investigated their loading characteristics. We observed the different kinds of losses that can be caused in the core and what is responsible for each of these losses. When the transformer is just powering up, there is an energizing inrush current which far exceeds that of the rated current for the transformer. This current leads to saturation and we also examined its impact. Finally, we compared the results from the measured and equivalent circuit over a variety of measured and came to the conclusion that this is a good approximation of the physical machine.